9.5 Making Change

WORKSHEET

The question

STEP I 1997, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q1

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself How many ways are there to make n pence using only 2p and 1p coins? Why does the answer depend only on how many 2p coins you use?

STAGE 1 Ten pence

Ask yourself Organise the count by the largest coins. How many 10p coins can you use? Given that, how many 5p coins?

STAGE 2 Building up layer by layer

Ask yourself Let $b(n)$ be the number of ways to make n pence from 5p, 2p and 1p coins. How can you express $b(n)$ using the function a?

STAGE 3 Twenty pence

Pause and try Notice the structure: each new coin adds a layer, and each layer is a sum of the previous layer at spacings of the new coin's value. Can you see how a computer would carry out this calculation for £1, or £100?

Try this next

1 **Two kinds of six.** (From STEP II, 2012.) Show that the coefficient of x^{24} in $(1-x^6)^{-2}(1-x^3)^{-1}$ is 15, and find the coefficient of x^n in general. Then show that the coefficient of x^{24} in $(1-x^6)^{-2}(1-x^3)^{-1}(1-x)^{-1}$ is 55, and find the coefficients of x^{25} and x^{66}. Describe a coin-counting problem that these coefficients answer.

2 **Distinct and odd.** Show that $(1+x)(1+x^2)(1+x^3)\cdots$ is the generating function for partitions into distinct parts. Use $1+x^k = \frac{1-x^{2k}}{1-x^k}$ to show that it equals $\frac{1}{(1-x)(1-x^3)(1-x^5)\cdots}$, and deduce Euler's theorem. Check it for $n=6$.

3 **A formula.** Show that the number of ways to make n pence using only 1p, 2p and 5p coins is the nearest integer to $\frac{(n+4)^2}{20}$. (Check it first for small n.)

4 **Order matters.** In how many ways can you make n pence by putting 1p and 2p coins into a slot one at a time, if the order matters? (You have met this sequence before.)

Further reading

- **George Andrews & Kimmo Eriksson, *Integer Partitions*** (Cambridge University Press). A short, readable introduction for undergraduates.
- **George Pólya, "On picture-writing"**, *American Mathematical Monthly* 63 (1956). Pólya's classic explanation of generating functions using the problem of changing a dollar (there are 292 ways).
- **Robert Kanigel, *The Man Who Knew Infinity*** (Abacus). The life of Ramanujan, and his partnership with Hardy.

The full guided solution, and what the question leads to at university, is in Section 9.5:
https://beyondstep.keithmcnulty.org/ch09/making-change.html